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Supplementary information

Three-photon microscopy: an emerging technique for deep intravital brain imaging

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Supplementary Information for

Three-photon microscopy: an emerging optical technique for deep intravital brain imaging

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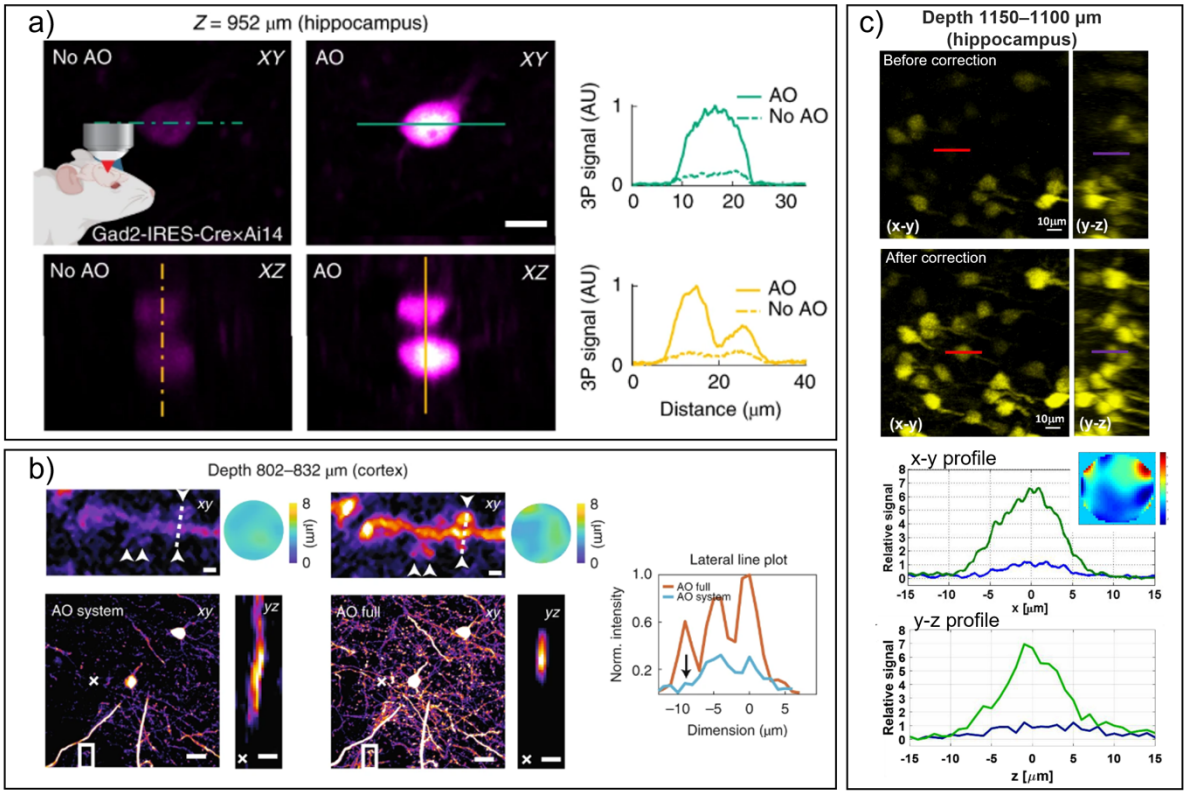


Fig. S1. Examples of three-photon imaging in deep brain tissue enhanced by adaptive optics. **a)** Lateral and axial images of neurons in the mouse hippocampus (Gad2-IRES-Cre × Ai14 (Rosa26-CAG-LSL-tdTomato)), 952 µm below dura, under 1,700 nm excitation, without and with AO (phase modulation and signal profiles along the green and yellow lines. Post objective power: 30 mW. Adapted from *Rodríguez C, et al., 2021, Nat Methods*. **b)** Images showing dendrites and spines in layer VI of the in vivo mouse brain (EGFP-Thy1(M)) through a cranial window recorded with wavefront correction of system aberrations (AO system) and with wavefront correction of system and brain tissue aberrations (AO full). The images show a maximum intensity projection and the plot on the right shows the intensity profile along spines (white dotted line). Scalebar is 20 µm and 2 µm on zoomed regions (corresponding to the white rectangle in the bottom images). Adapted from *Streich, L. et al., 2021, Nat. Methods*. **c)** Maximal-intensity projections of YFP-labeled hippocampus neurons before and after correction. The graphs show the signal improvement of the corrected image (green) compared with the non-corrected image (blue) both in x-y and y-z profiles. Insert shows the phase pattern applied with the SLM for the correction. Adapted from *Sinefeld D, et al., 2022, Front. Neurosci.*

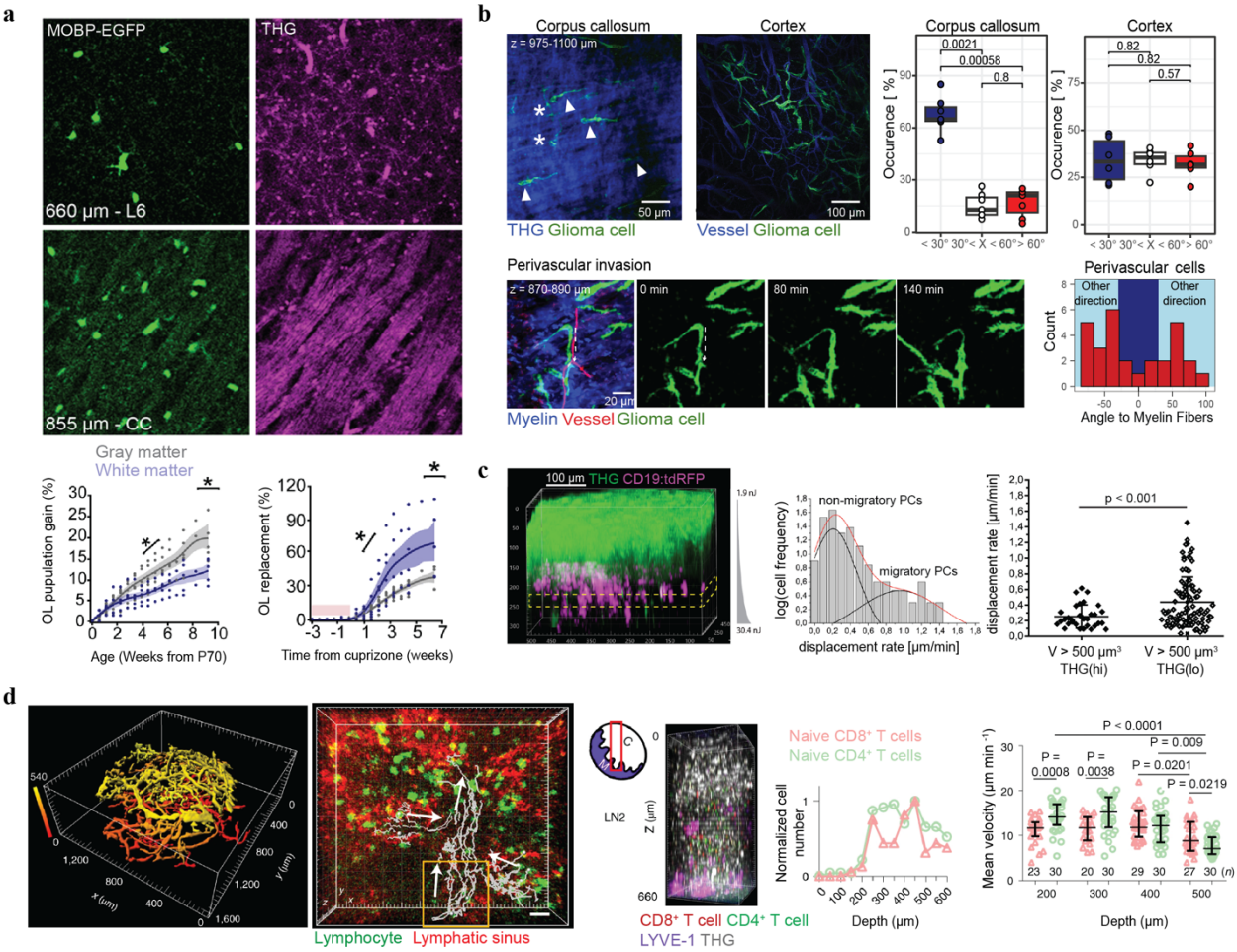


Fig. S2. Example studies employing 3PM for biological questions. a) *Top:* 3PM images of mature oligodendrocytes (OL) in MOBP-EGFP mice (left) and THG signal of vessels/myelinated fibers in cortex layer 6 (top) and corpus callosum (bottom). *Bottom (left):* OL population growth comparison between gray and white matter, showing a significant increase in gray matter. *Bottom (right):* OL repopulation after cuprizone-induced demyelination, favoring white matter. *(Adapted from Thornton et al., Nature Neuroscience 2024).* **b)** *Top (left):* Multiphoton images of S24 Glioblastoma cells (GBCs) in CC (3PM) and cortex (2PM) in a PDX mouse model, showing parallel (<30°) invasion along myelinated fibers in CC and undirected growth in cortex. Arrows: exemplary GBCs; Asterisks: orthogonal invasion. *Bottom (left):* Example of perivascular GBC invasion non-parallel to fibers. *Bottom (right):* Angle analysis reveals increased non-parallel invasion in perivascular GBCs in CC. *(Adapted from Schubert et al., Nature Communications 2024).* **c)** *Left:* 3D 3PM image of intact mouse tibia (1650 nm, 3 MHz, pulse-energy 1.9–30.4 nJ). Green: THG signal; Magenta: CD19:tdRFP+ plasma/B cells distinguished by volume. *Middle:* In vivo 3PM timelapse shows two plasma cell populations, with displacement rates forming a double Gauss-peak distribution. *Right:* Comparison of THG(hi) and THG(lo) plasma cells, where THG(hi) indicates an abundance of ER, associated with higher antibody production. *(Adapted from Rakhymzhan et al.,* iScience *2024).* **d**) Left: Depth-color map of fluorescein+ blood vessels in a popliteal lymph node. Middle: Migration of eGFP+ lymphocytes in a lymphatic sinus (red = LYVE-1-eFluor615+ endothelial cells), with flow direction shown by white arrows. Right: CD8+ and CD4+ T cell velocities at different depths, showing depth-dependent CD4+ T cell movement. (Adapted from Choe et al., Nature Immunology 2022).

SI Table 1: Overview of published data on several fluorophores available for 3P excitation including fluorescent proteins, organic dyes and nanoparticles. 3P absorption cross-sections (σ^3) are given in 10^{-84} cm^6 $(s/photons)^2$, the column named 3P spectrum indicates whether the 3P absorption spectrum is available around the wavelengths of interest. The column named S ~ P^3 indicates whether the cubic dependence of the fluorescence signal on the excitation power has been verified.

	Fluorophore	σ^3 at 1300 nm	σ^3 at 1700 nm	3P spectrum	S ~ P^3	Reference
Fluorescent Proteins	wGFP	15.9	-	NO	YES	[1]
	GCaMP6f +Ca	180	-	YES	NO	link
	GCaMP6f -Ca	15	-	YES	NO	link
	mBanana	-	1.8	YES	NO	[2]
	mOrange	-	10	YES	NO	[2]
	dsRed	-	14	YES	NO	[2]
	mStrawberry	-	34	YES	YES	[2]
	mRaspberry	-	-	YES	NO	[3]
	mRaspberry	-	48	YES	NO	[2]
	mPlum	-	9	YES	NO	[2]
	mKate	-	240	YES	YES	[2]
	mCherry	9	90	YES	NO	[4]
	mCherry	-	-	YES	NO	[3]
	dsRed	13	-	YES	NO	[4]
	dsRed	-	-	NO	YES	[5]
	tdTomato	-	56	YES	YES	[6]
	tdTomato	230	-	YES	NO	[4]
	tdTomato	-	-	YES	NO	[3]
	RFP	-	-	NO	YES	[7]
	eRFP	-	25	YES	NO	[2]
dyes	Fluorescein	16.3	-	NO	YES	[1]
	Fluorescein	-	-	YES	NO	[3]
	Oregon Green 488 BAPTA-1 hexapotassium salt	-	-	YES	NO	[3]
	red algae RFP	-	410	YES	YES	[6]
	Sulforhodamine 101	-	-	NO	YES	[1]
	Sulforhodamine 101	-	-	YES	NO	[3]
	Sulforhodamine 101	$1.4*10^3$	40	YES	NO	[4]
	Sulforhodamine 101	-	47.9	YES	NO	[8]
	Texas Red	-	-	YES	NO	[3]
	Texas Red	$1.1*10^3$	56	YES	YES	[4]
	Texas Red	-	61	YES	NO	[8]
	Alexa Fluor 546	120	24	YES	NO	[4]

	Alexa Fluor 633 hydrazide	-	-	YES	NO	[3]
	Alexa Fluor 647	-	-	NO	YES	[5]
nanoparticles	Qdot 605	10^7	$1.5*10^6$	YES	NO	[4]
	TPATCN@F-127	$1.93*10^6$	-	YES	YES	[9]
	CdTe/CdSe QDs	-	-	YES	YES	[10]
	Qtracker655	-	$5.51*10^6$	YES	YES	[8]
	Qtracker705	-	$2.74*10^6$	YES	NO	[8]
	Qtracker800	-	$4.18*10^6$	YES	NO	[8]
	Beida630	-	$1.4*10^5$	YES	NO	[8]

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